Activity and Sequence Diagrams

Ten detailed use cases, chosen to cover every actor of the system. Each use case has one UML activity diagram (with swimlanes) and one UML sequence diagram that follows the adopted Modular Monolith architecture.

#	Use case	Name	Primary actor	Requirements (SRS v3.0)	Pages
1	**UC-01.2**	Log In	Registered User (any role)	FR-ACC-02, FR-ACC-04, BR-13	3–4
2	**UC-02.3**	Filter Projects	Broker	FR-FLT-02 to FR-FLT-09, FR-MAP-04, BR-03, BR-08, BR-09	5–6
3	**UC-02.8**	Reveal Sales Contact	Broker	FR-PRJ-07, FR-CNT-01 to FR-CNT-03, BR-10	7–8
4	**UC-03.5**	Generate Catalog PDF	Broker	FR-CAT-04 to FR-CAT-06, BR-11, BR-12 (includes UC-03.6)	9–10
5	**UC-04.3**	Invite Member	Agency Admin	FR-AGY-02, FR-ADM-09, FR-ACC-01 (includes UC-04.4)	11–12
6	**UC-04.7**	Deactivate or Reactivate Member	Agency Admin	FR-AGY-04, FR-ACC-04	13–14
7	**UC-05.5**	Publish Project	Data Operator	FR-DAT-04, FR-DAT-05, BR-08 (includes UC-05.6)	15–16
8	**UC-05.9**	Manage Units (change price or status)	Data Operator	FR-DAT-08, FR-DAT-09, BR-01, BR-16	17–18
9	**UC-06.2**	Suspend or Reactivate Agency	Platform Admin	FR-ADM-02, FR-ADM-08, BR-13	19–20
10	**UC-06.5**	Manage Exchange Rates	Platform Admin	FR-ADM-05, FR-LOC-01, BR-03	21–22

Coverage by actor

- **Registered User (any role)**: UC-01.2
- **Broker**: UC-02.3, UC-02.8, UC-03.5
- **Agency Admin**: UC-04.3, UC-04.7
- **Data Operator**: UC-05.5, UC-05.9
- **Platform Admin**: UC-06.2, UC-06.5

How to read the diagrams

Activity diagrams

- **Swimlanes** show who performs each action: the actor, the web application (Next.js), the Backend API (NestJS), and, where used, the PDF Worker, the database or the email service.
- ● initial node · ◉ activity final · ⊗ flow final (this path ends, the activity does not).
- ◇ decision / merge. Guards are written in brackets, for example *[yes]*.
- ▬ fork / join: the paths between them run in parallel (catalog generation).
- Notes cite the business rule or requirement that drives a step.

Sequence diagrams

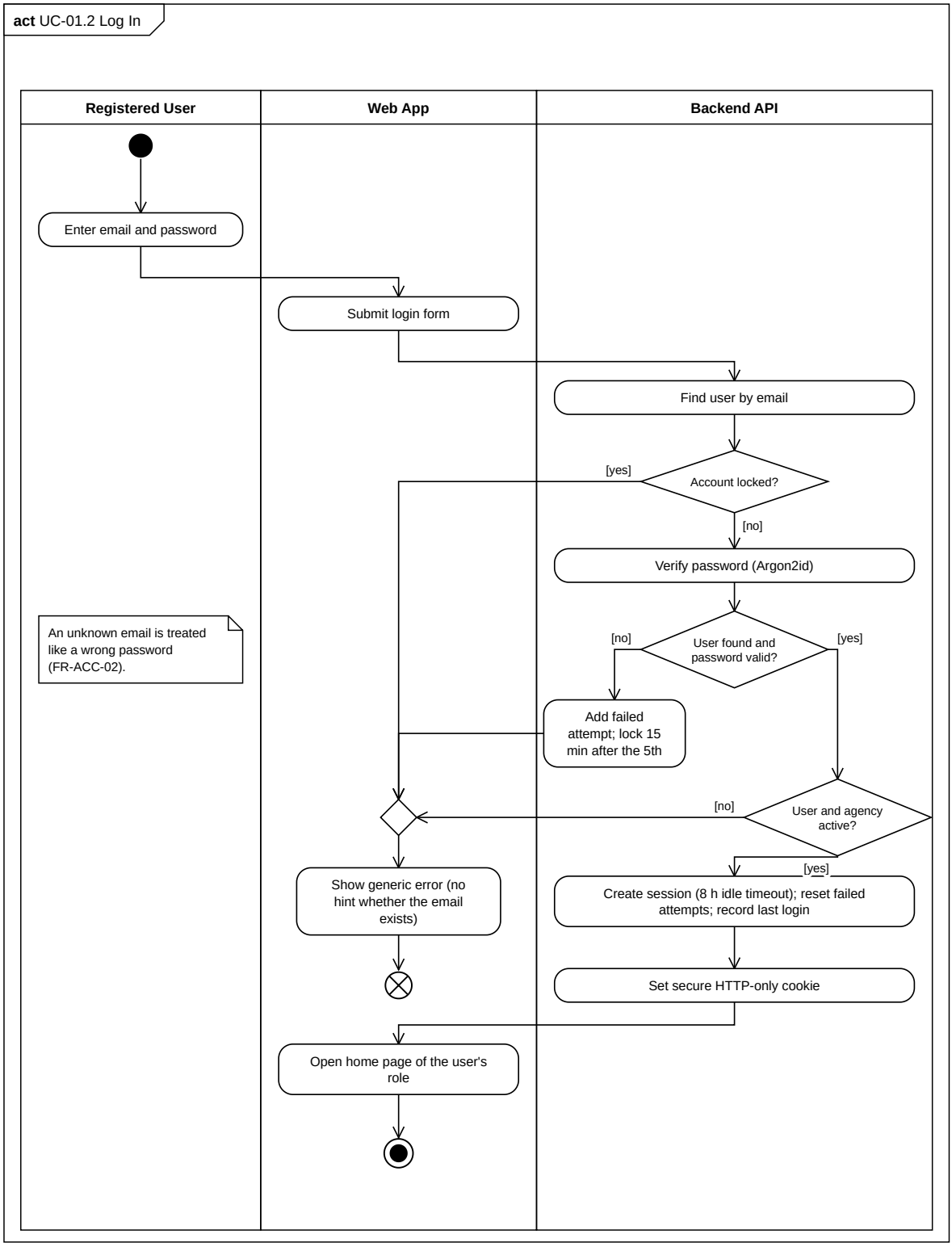
- Lifelines follow the adopted architecture: **«boundary»** a page of the web app; **«control»** a controller, service or worker inside a module; **«module export»** the public interface of another module (modules never read each other's tables); **«entity»** a repository or domain object; **«database»**, **«queue»**, **«external»** infrastructure.
- Solid line with filled head: synchronous call. Solid line with open head: asynchronous message (queue, event). Dashed line: return.
- Combined fragments: **alt** alternatives, **opt** optional, **loop** repetition, **ref** another use case (included).
- Messages are numbered in order; returns are not numbered.

Sources: REIH SRS v3.0 (functional requirements, business rules, use case model) and the project's adopted architecture, ERD and class diagrams.

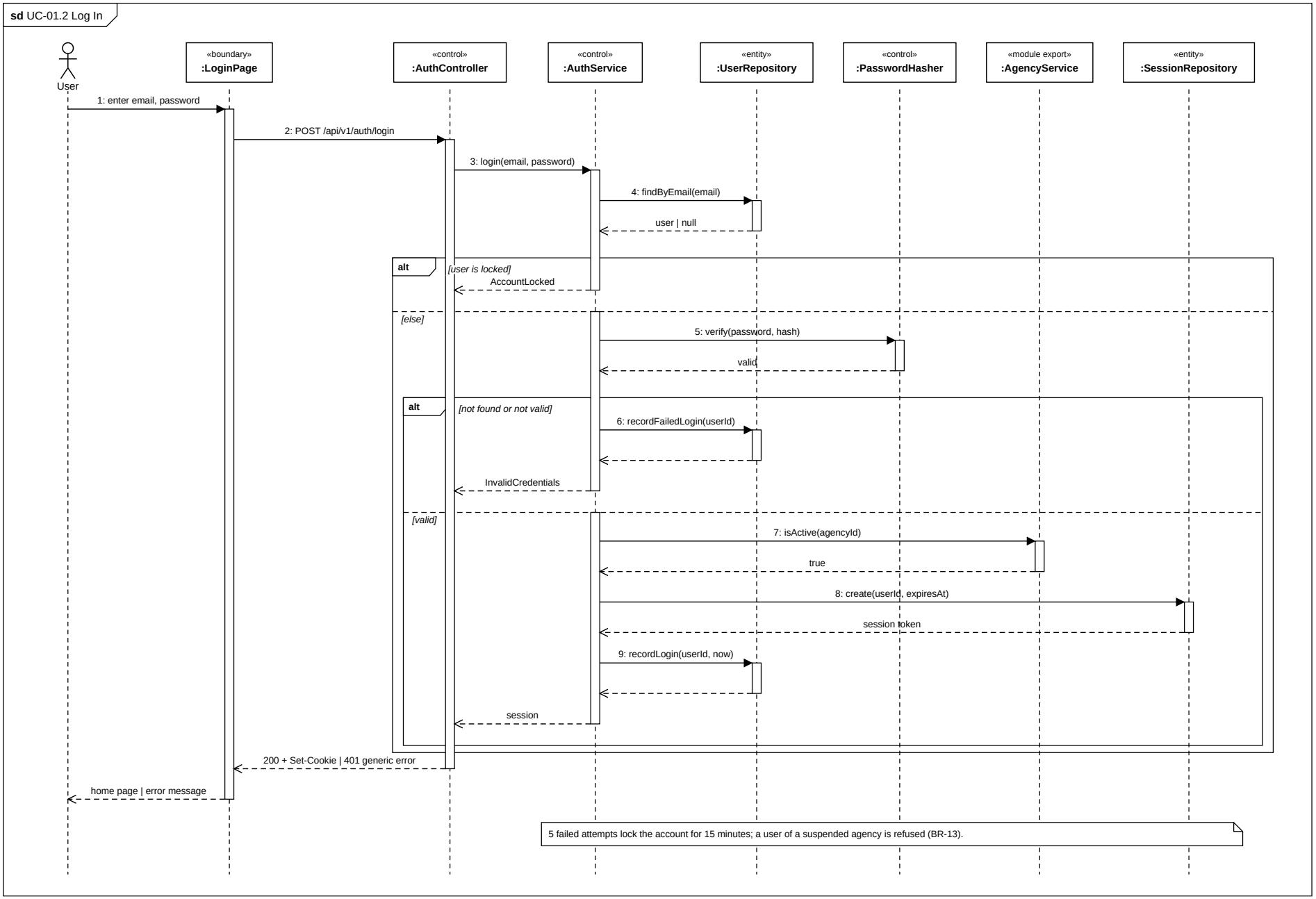
UC-01.2 Log In

Primary actor	Registered User (any role)	**Requirements**	FR-ACC-02, FR-ACC-04, BR-13
Description	A user signs in with email and password and receives a server-side session.		
Precondition	The account has been activated (UC-01.1).		
Postcondition	A session exists and a secure HTTP-only cookie is set; or the attempt is refused with a generic message.		

Activity diagram



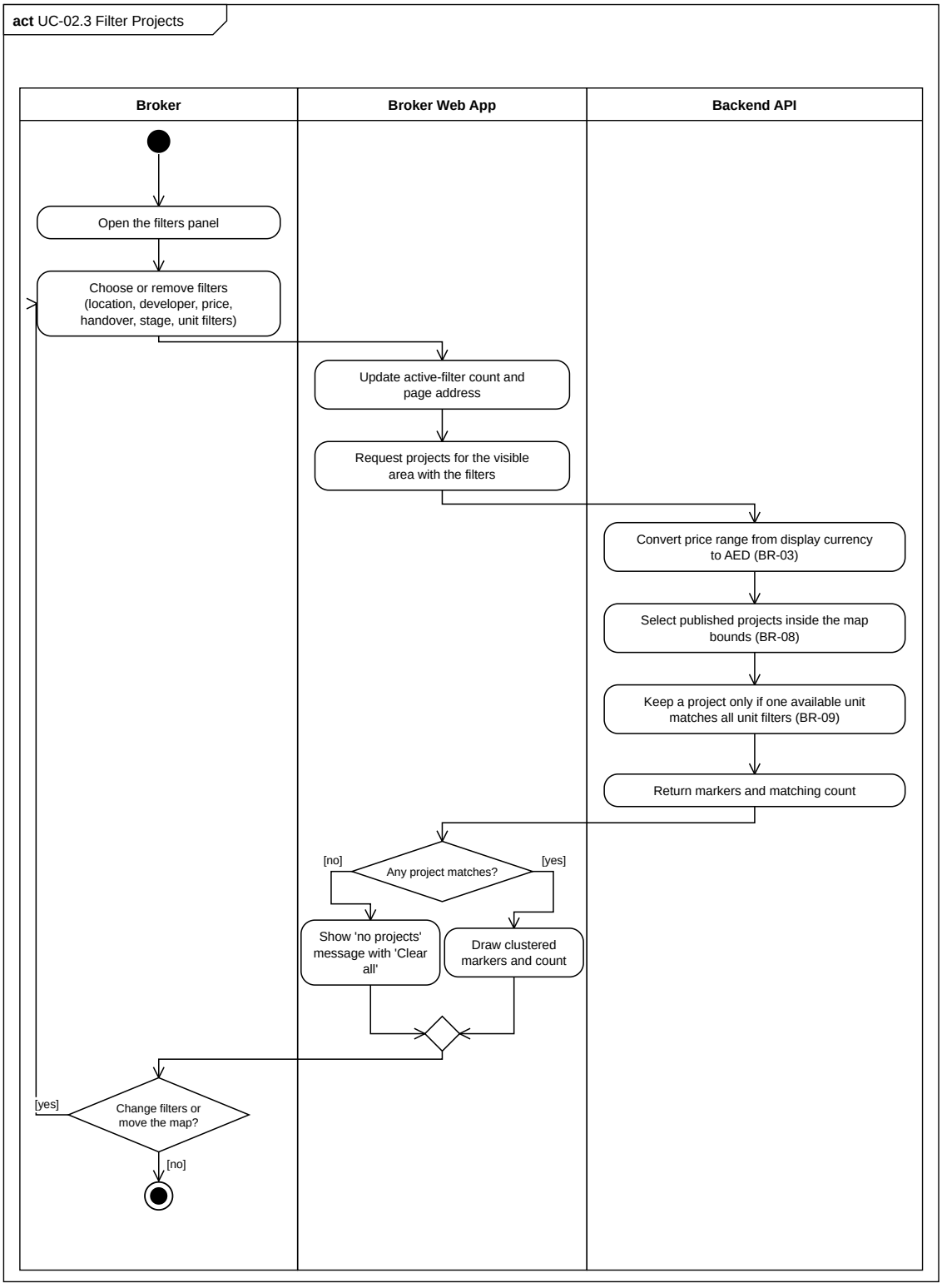
UC-01.2 Log In — Sequence diagram



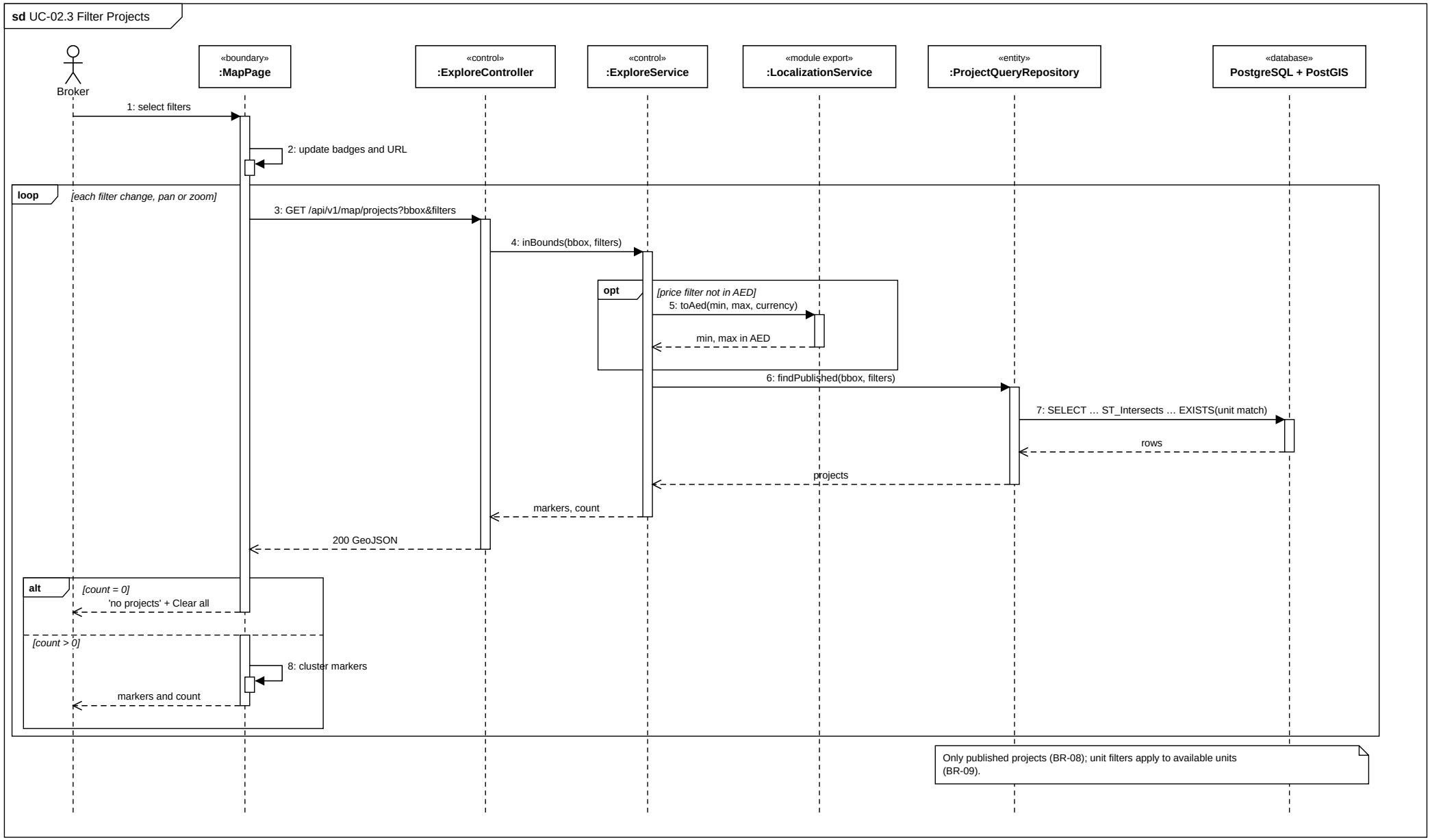
UC-02.3 Filter Projects

Primary actor	Broker	**Requirements**	FR-FLT-02 to FR-FLT-09, FR-MAP-04, BR-03, BR-08, BR-09
Description	The broker narrows the map to projects that match location, developer, price, handover, stage and unit filters.		
Precondition	The broker is logged in and the map is open (UC-02.1).		
Postcondition	The map and the result count show only matching published projects inside the visible area.		

Activity diagram



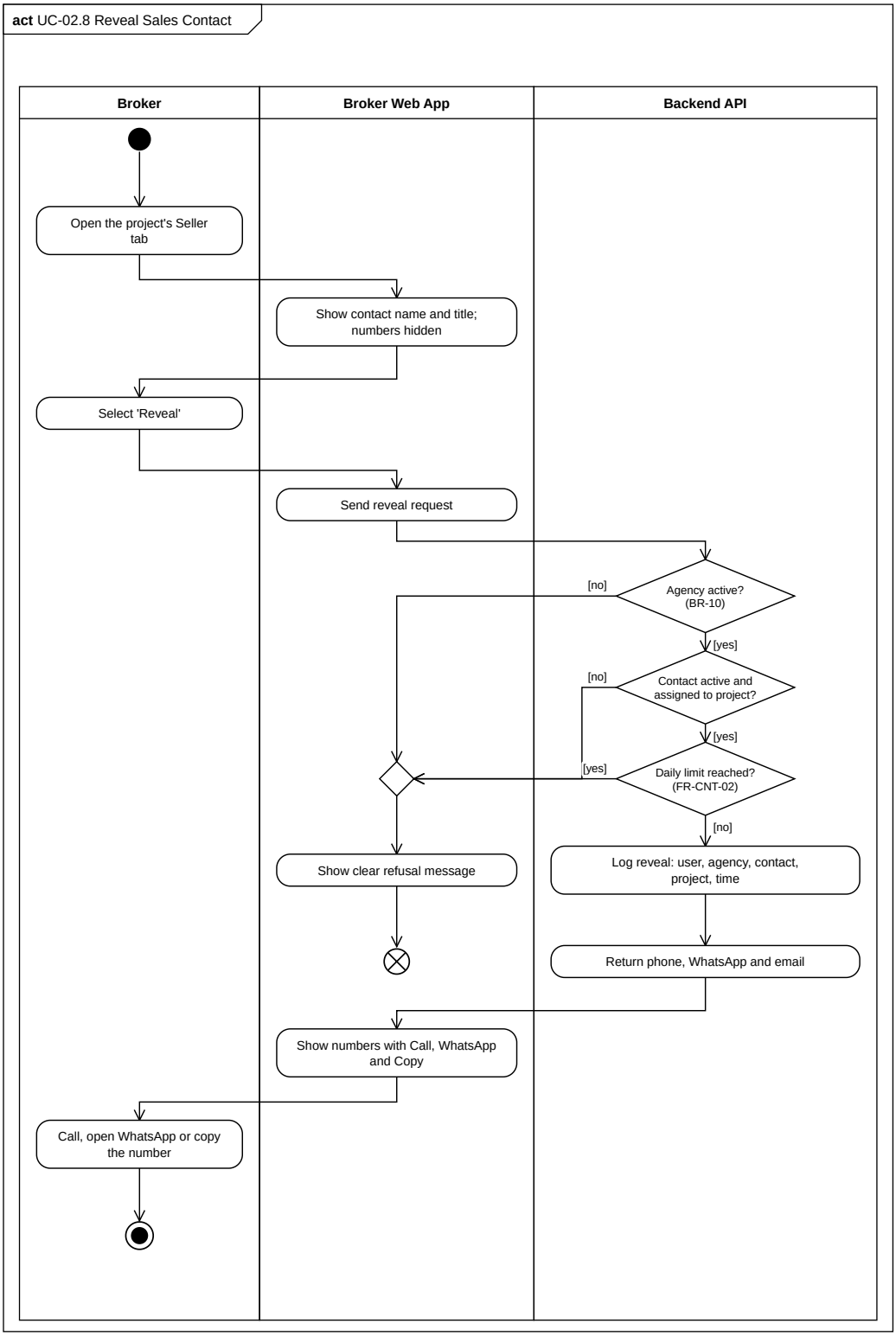
UC-02.3 Filter Projects — Sequence diagram



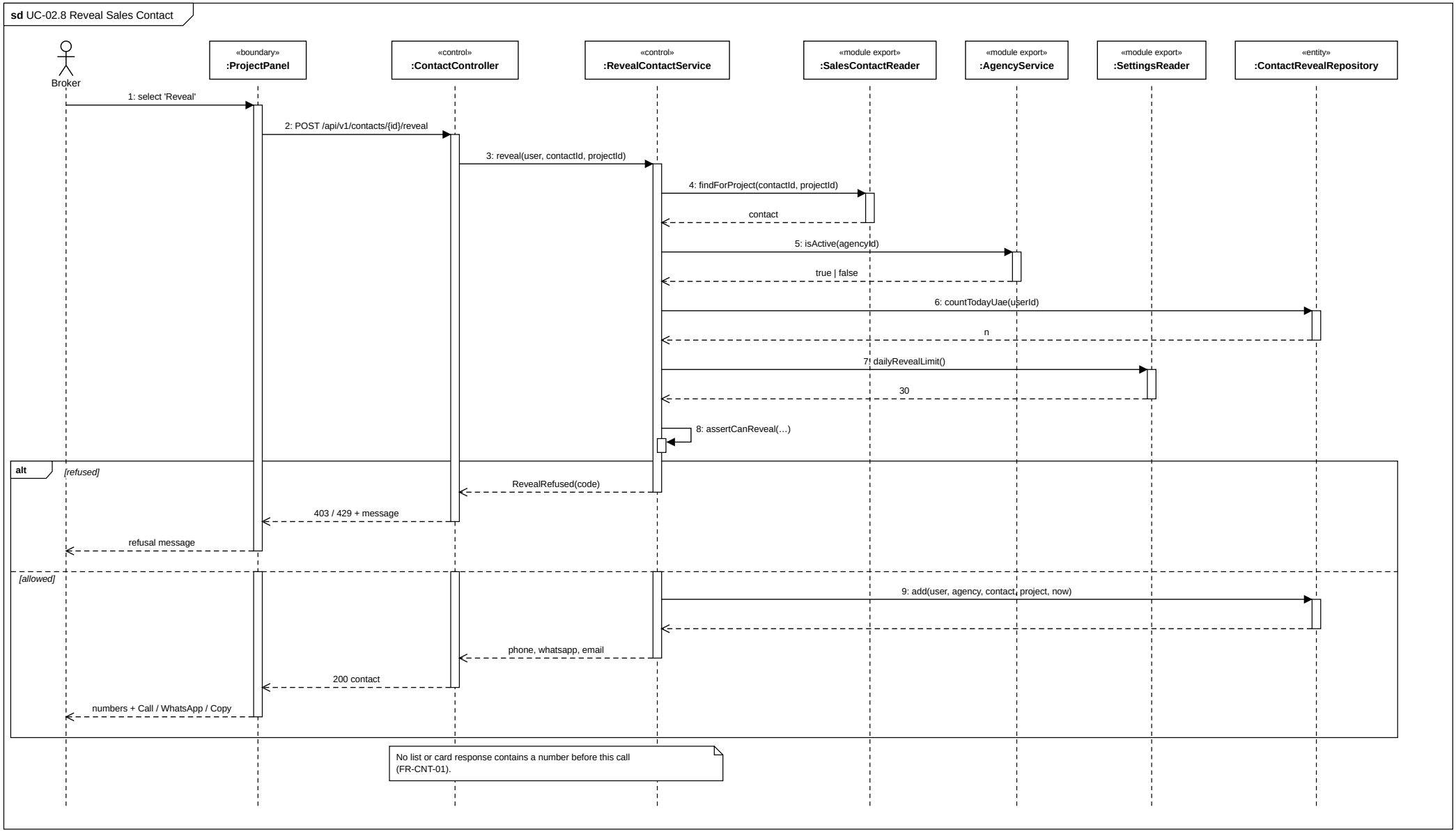
UC-02.8 Reveal Sales Contact

Primary actor	Broker	**Requirements**	FR-PRJ-07, FR-CNT-01 to FR-CNT-03, BR-10
Description	The broker reveals a developer sales manager's phone, WhatsApp and email from the Seller tab; every reveal is logged.		
Precondition	The broker is logged in and a published project card is open (UC-02.4).		
Postcondition	The numbers are shown with Call, WhatsApp and Copy, and the reveal is logged; or a clear refusal is shown.		

Activity diagram



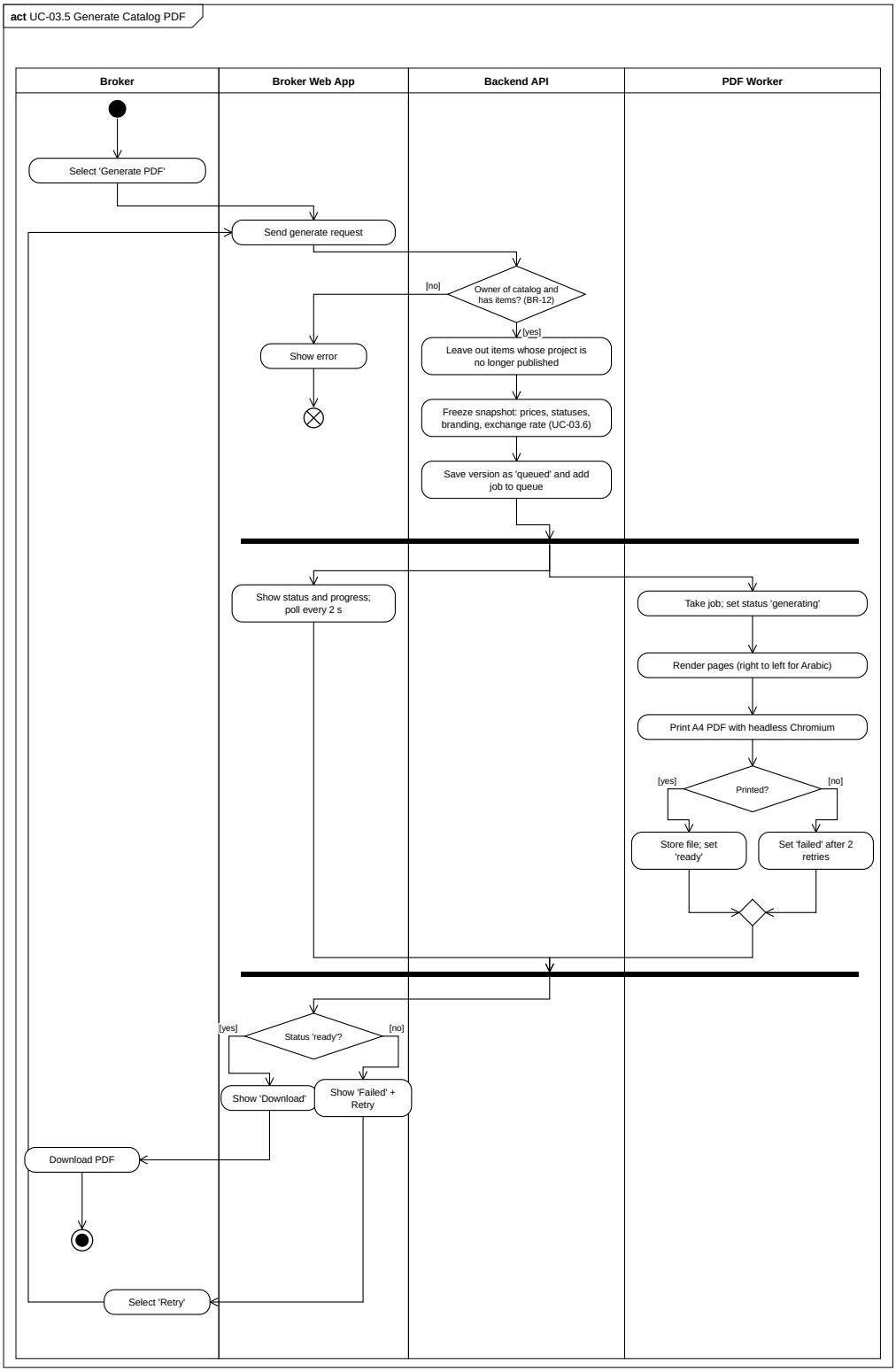
UC-02.8 Reveal Sales Contact — Sequence diagram



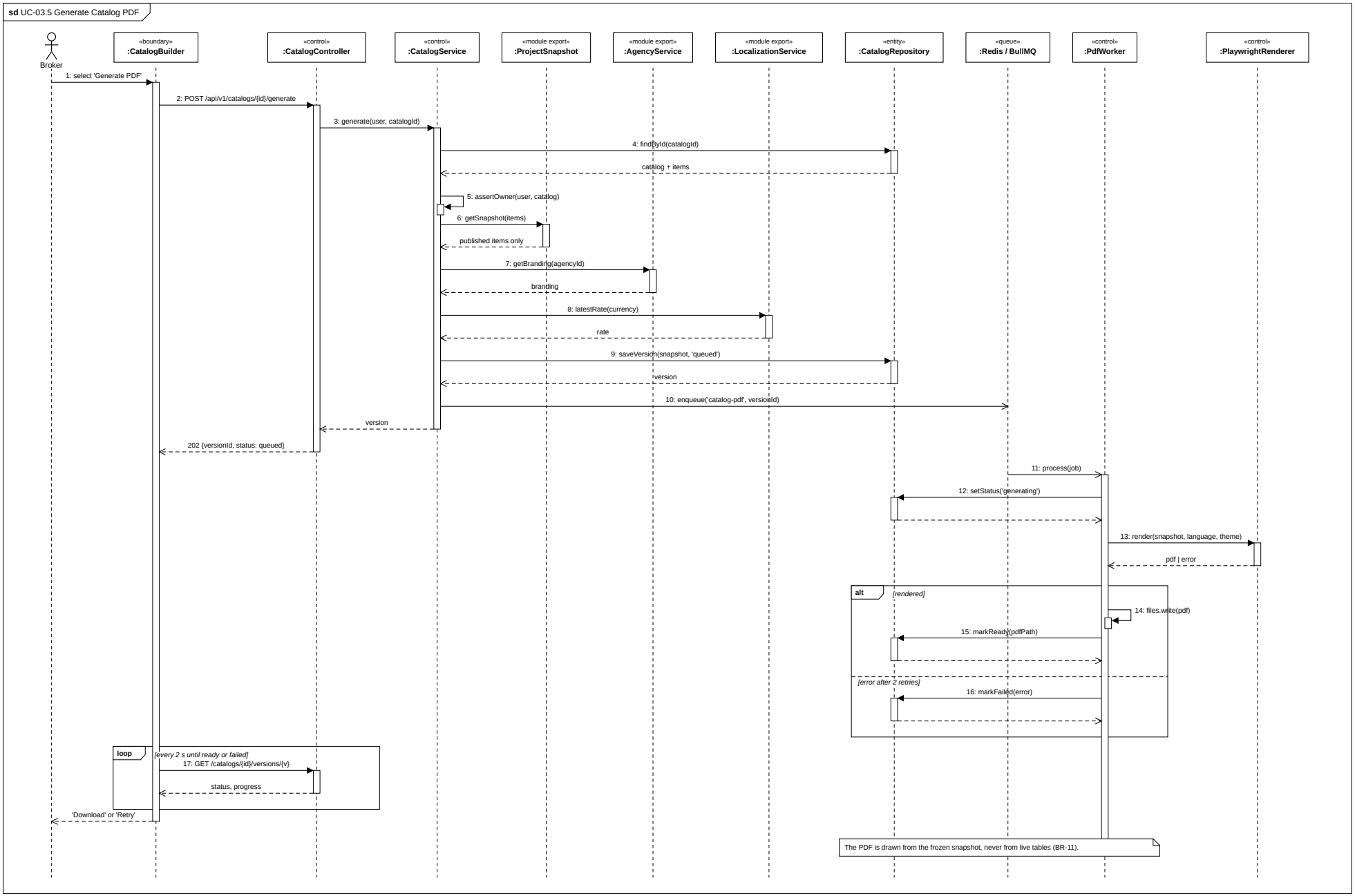
UC-03.5 Generate Catalog PDF

Primary actor	Broker	**Requirements**	FR-CAT-04 to FR-CAT-06, BR-11, BR-12 (includes UC-03.6)
Description	The broker generates a branded A4 PDF of a catalog. Generation runs in the background and produces a frozen version.		
Precondition	The broker owns the catalog and it contains at least one item.		
Postcondition	A new catalog version is 'ready' with its PDF, or 'failed' after two automatic retries.		

Activity diagram



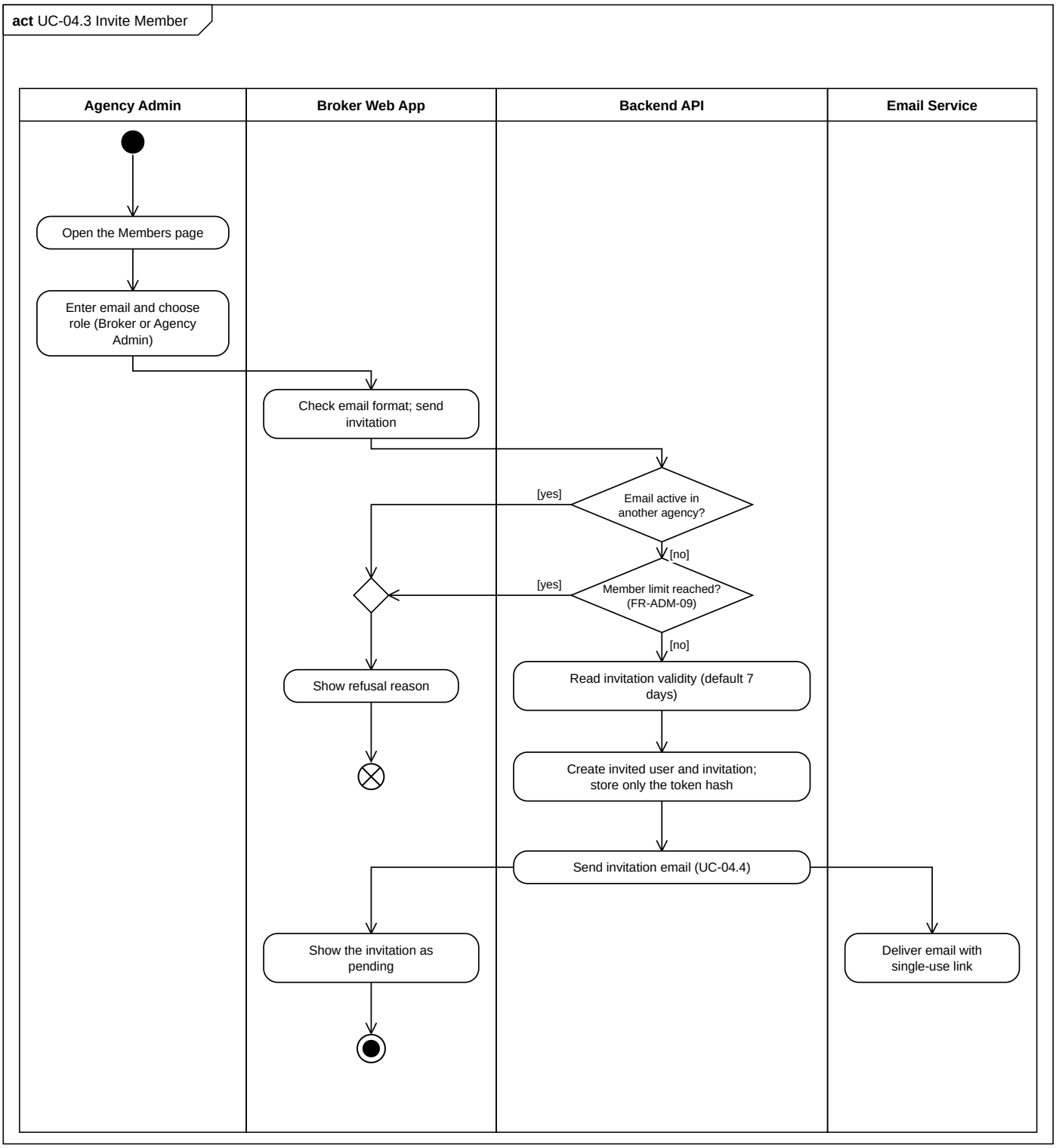
UC-03.5 Generate Catalog PDF — Sequence diagram



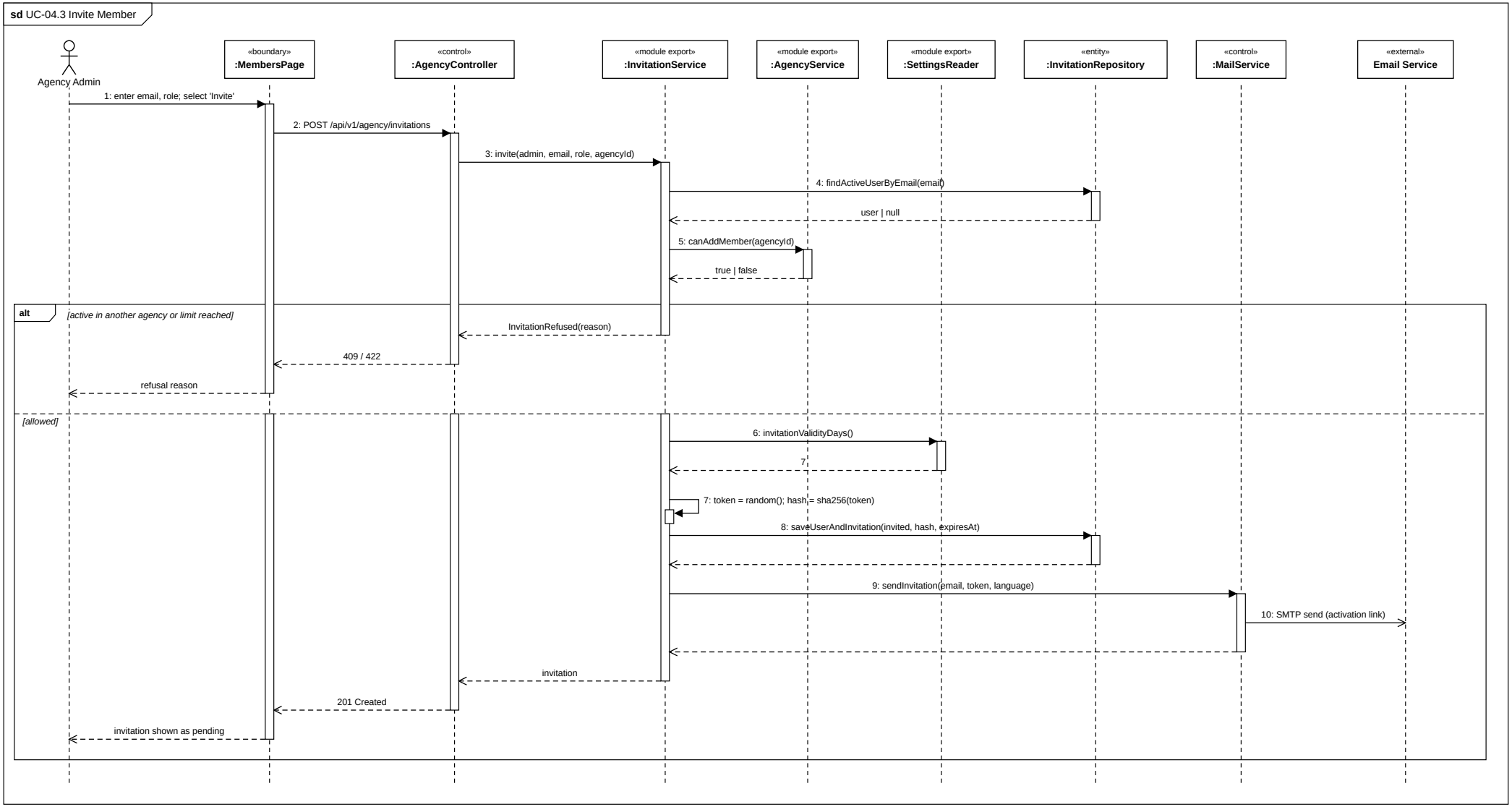
UC-04.3 Invite Member

Primary actor	Agency Admin	**Requirements**	FR-AGY-02, FR-ADM-09, FR-ACC-01 (includes UC-04.4)
Description	The Agency Admin invites a person by email as Broker or Agency Admin; the system emails a single-use activation link.		
Precondition	The Agency Admin is logged in and the agency is active.		
Postcondition	An invited user and a pending invitation exist and the email is sent; or the invitation is refused.		

Activity diagram



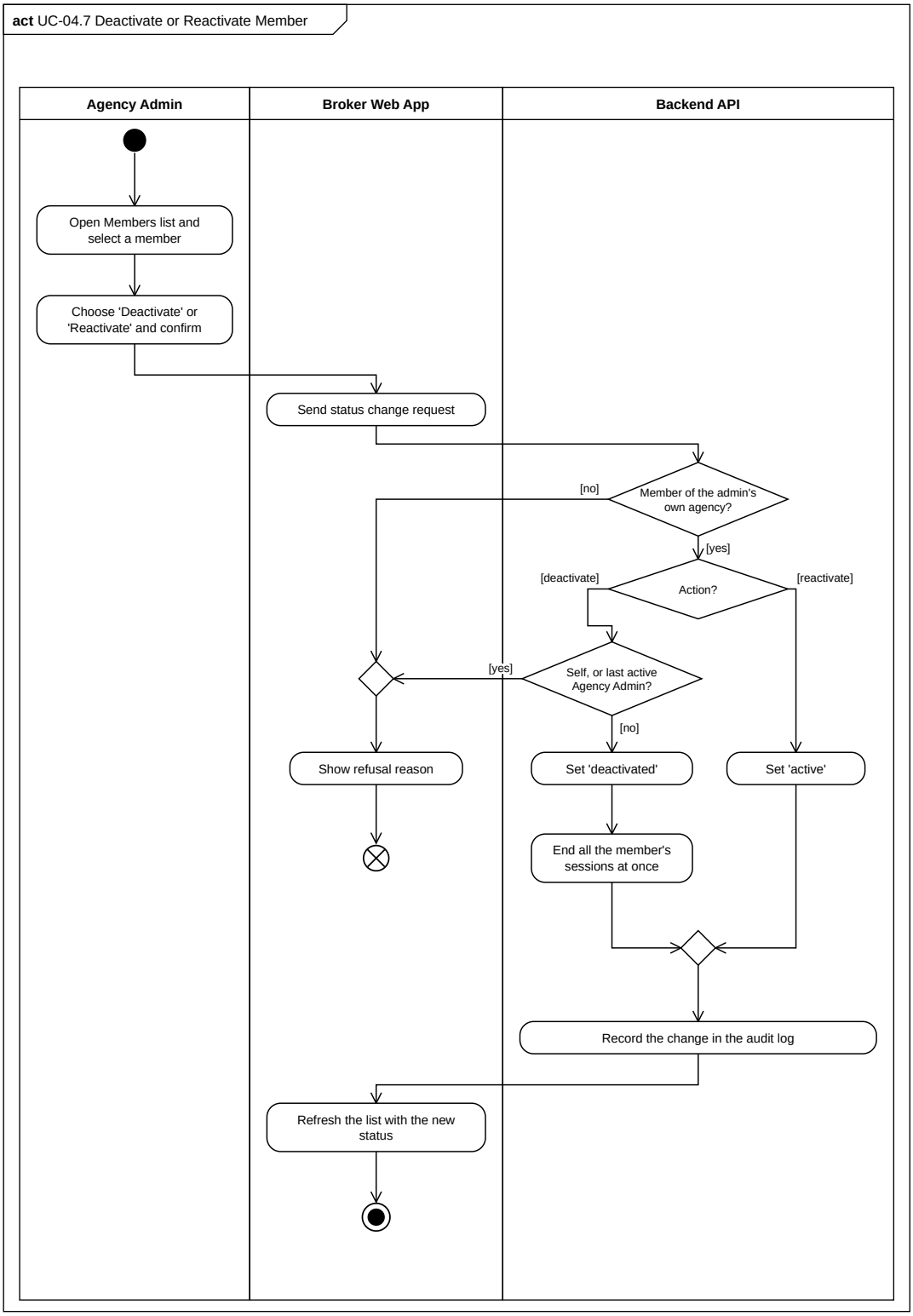
UC-04.3 Invite Member — Sequence diagram



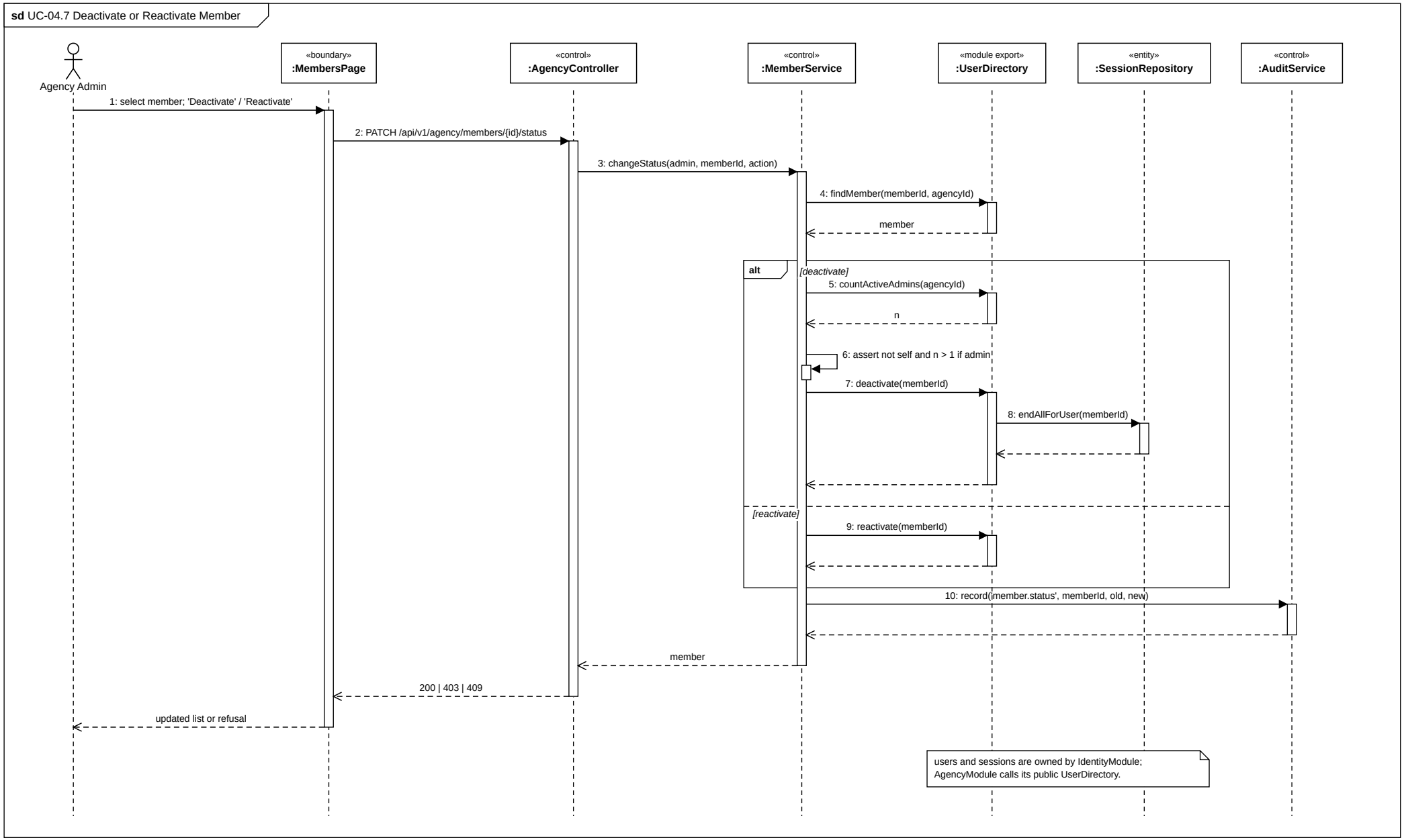
UC-04.7 Deactivate or Reactivate Member

Primary actor	Agency Admin	**Requirements**	FR-AGY-04, FR-ACC-04
Description	The Agency Admin deactivates a member (ending their sessions at once) or reactivates a deactivated member.		
Precondition	The Agency Admin is logged in; the member belongs to the same agency.		
Postcondition	The member's status is changed and, on deactivation, all their sessions are ended.		

Activity diagram



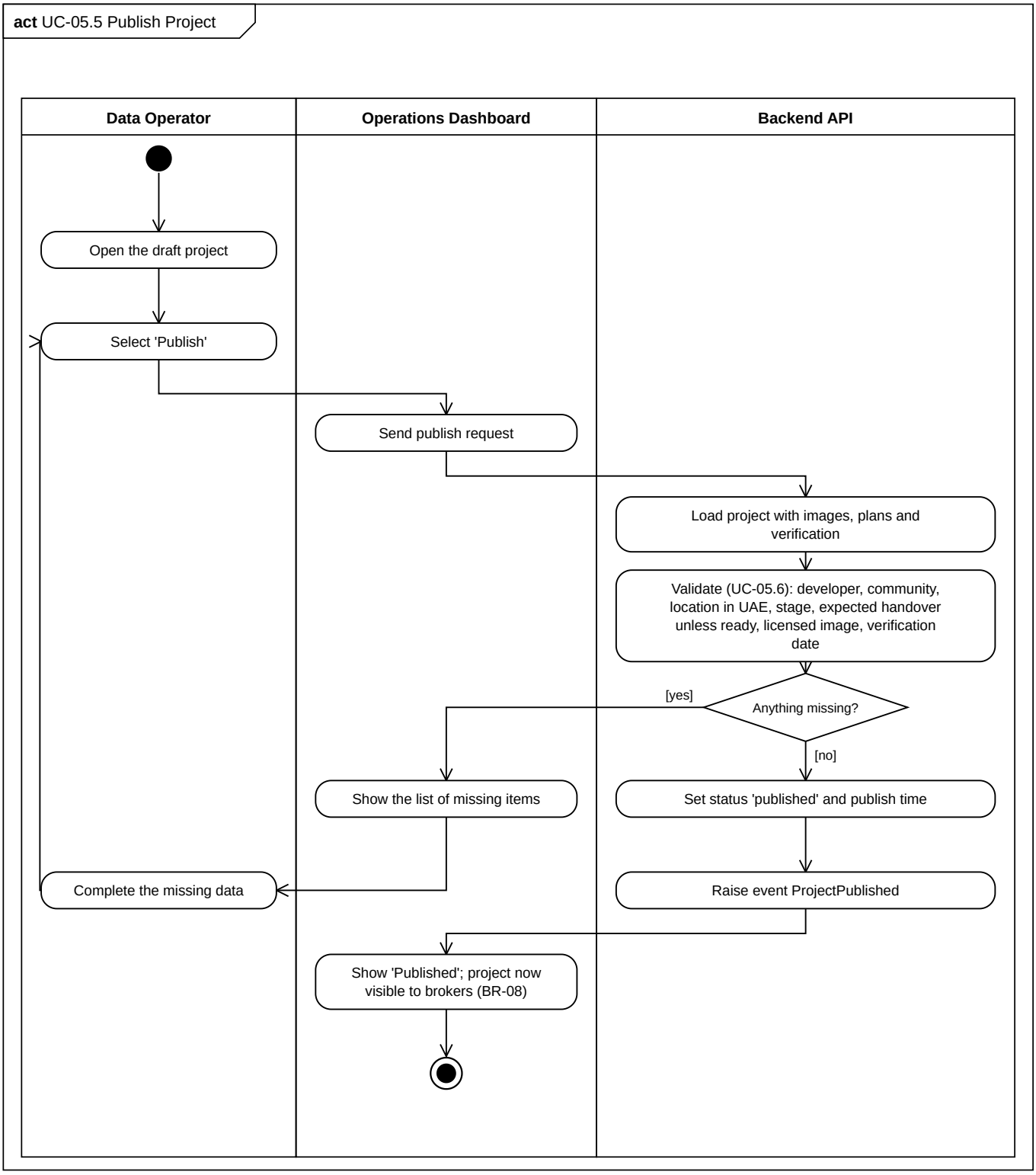
UC-04.7 Deactivate or Reactivate Member — Sequence diagram



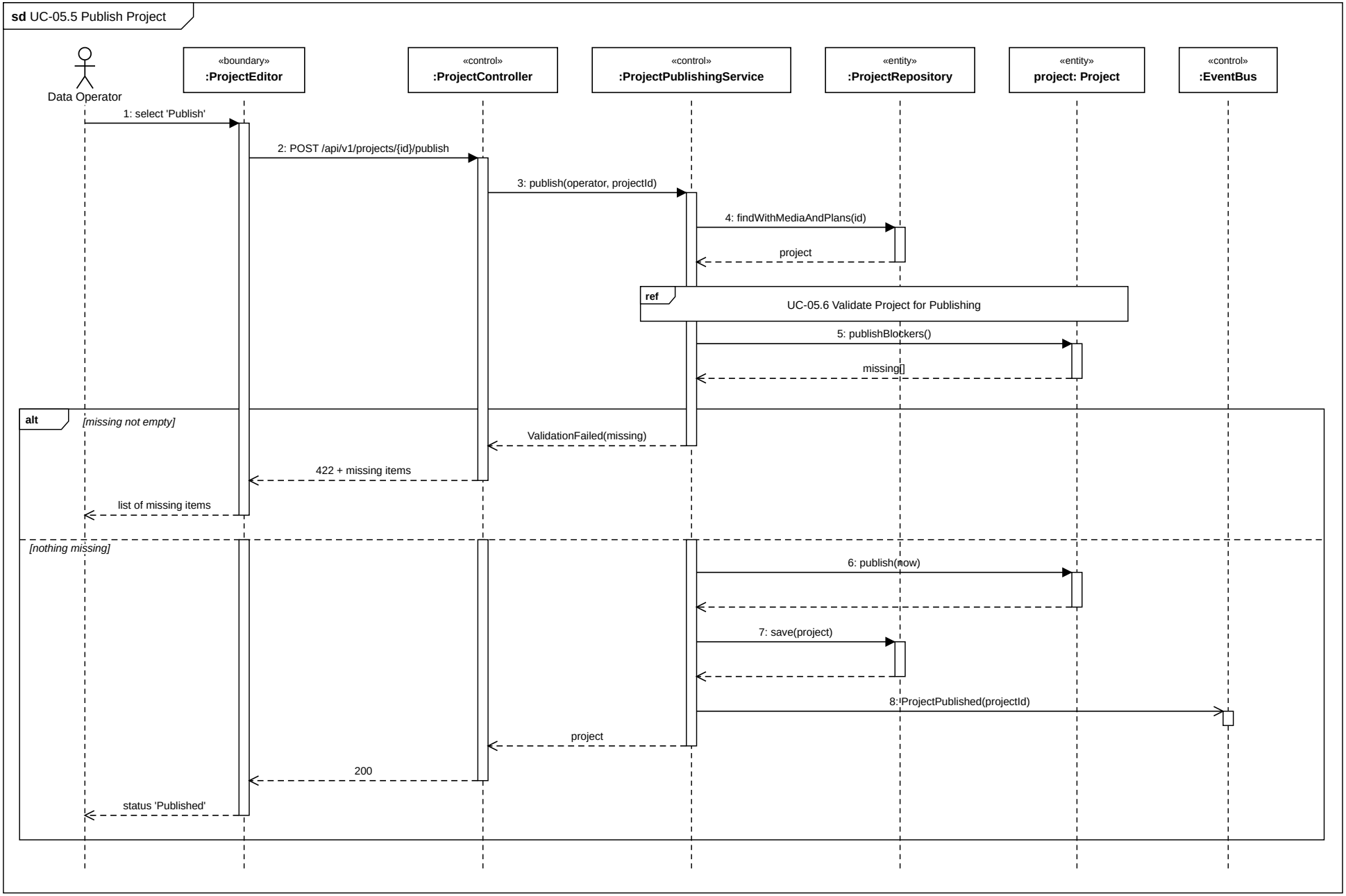
UC-05.5 Publish Project

Primary actor	Data Operator	**Requirements**	FR-DAT-04, FR-DAT-05, BR-08 (includes UC-05.6)
Description	The Data Operator publishes a draft project; the system checks the required data first and names anything missing.		
Precondition	The Data Operator is logged in and the project exists as a draft.		
Postcondition	The project is 'published' and visible to brokers; or it stays a draft and the missing items are listed.		

Activity diagram



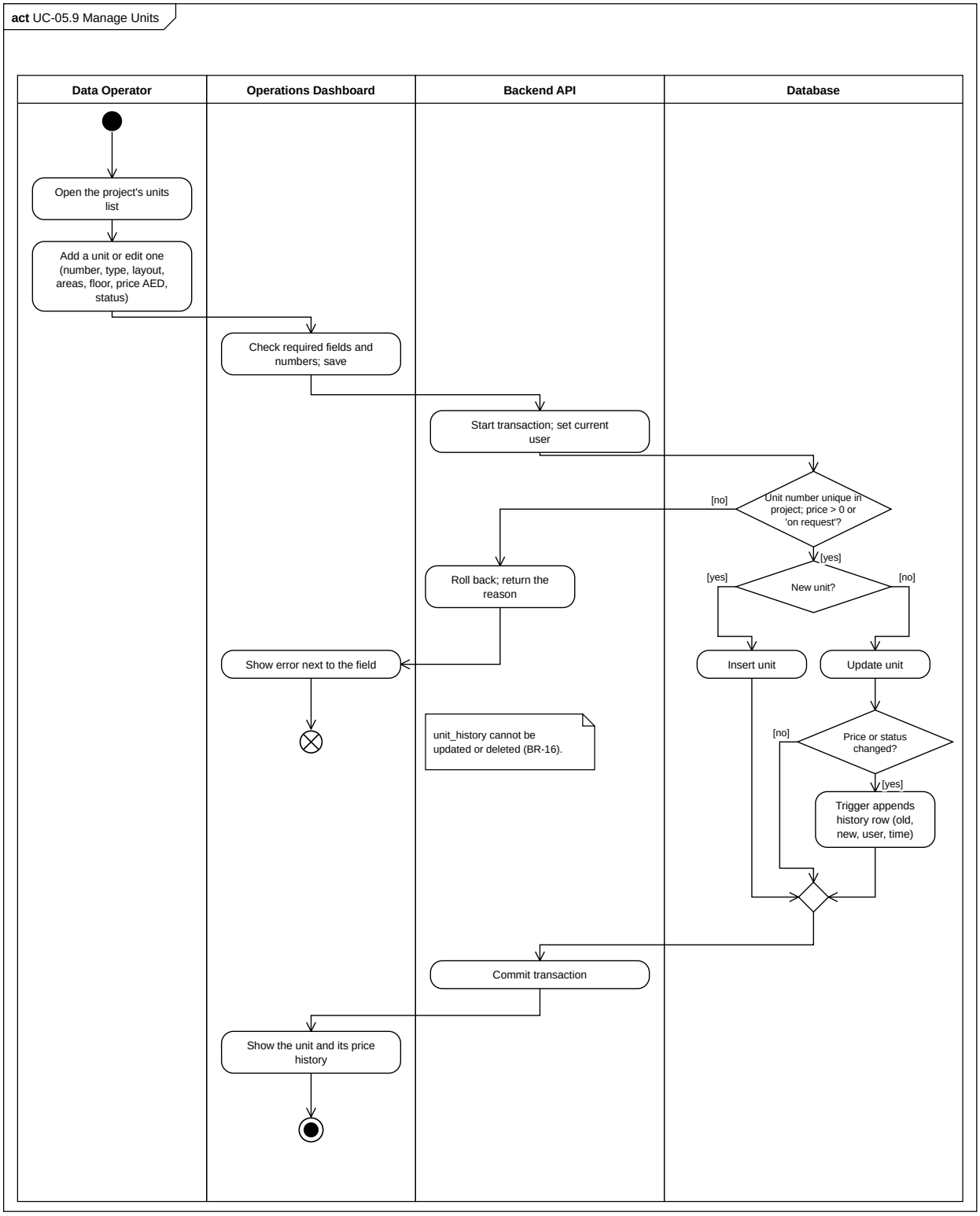
UC-05.5 Publish Project — Sequence diagram



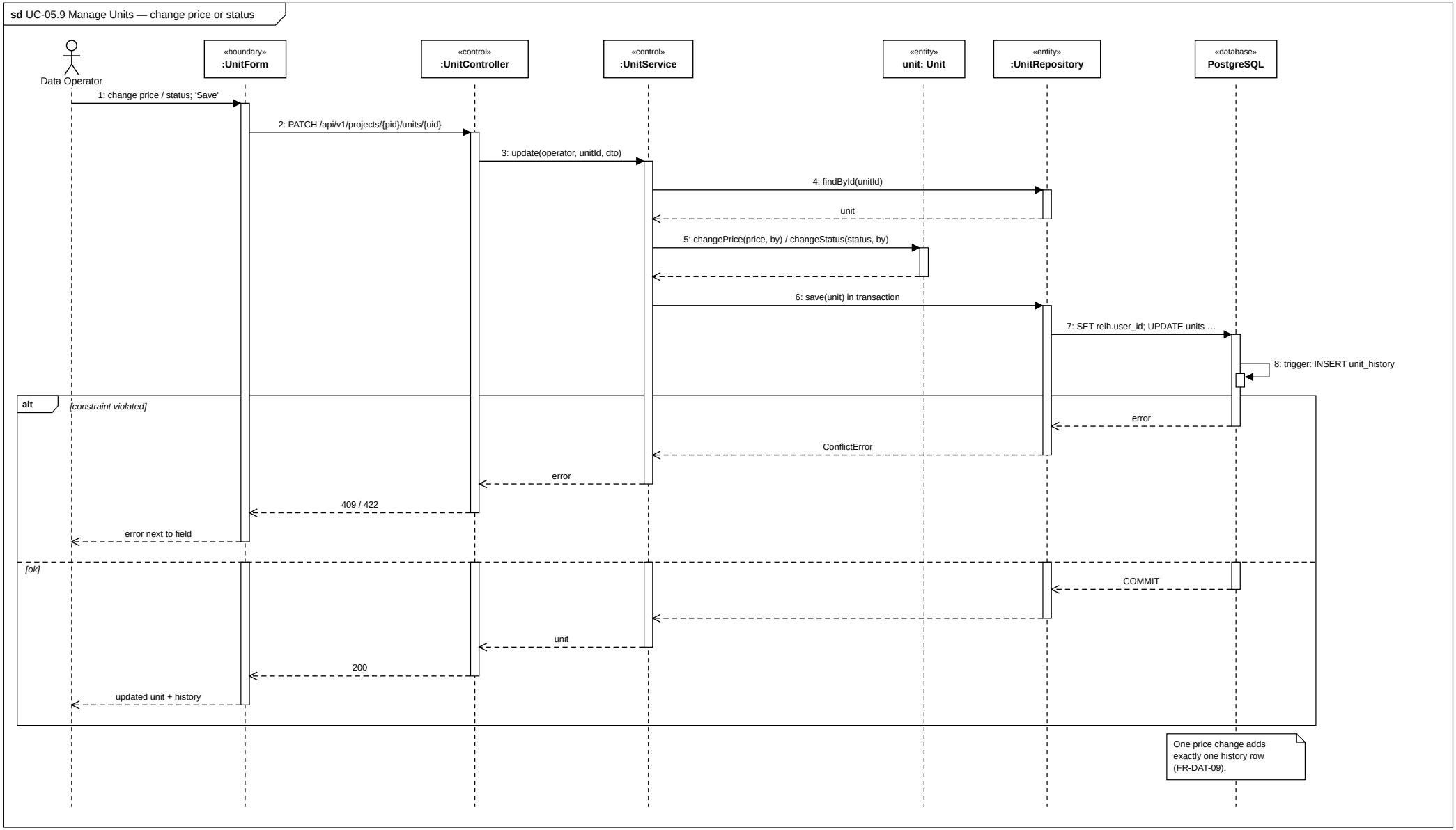
UC-05.9 Manage Units (change price or status)

Primary actor	Data Operator	**Requirements**	FR-DAT-08, FR-DAT-09, BR-01, BR-16
Description	The Data Operator adds or edits a unit. Every change of price or status is stored as an append-only history record.		
Precondition	The Data Operator is logged in and the project exists.		
Postcondition	The unit is saved; one history record exists for each changed price or status.		

Activity diagram



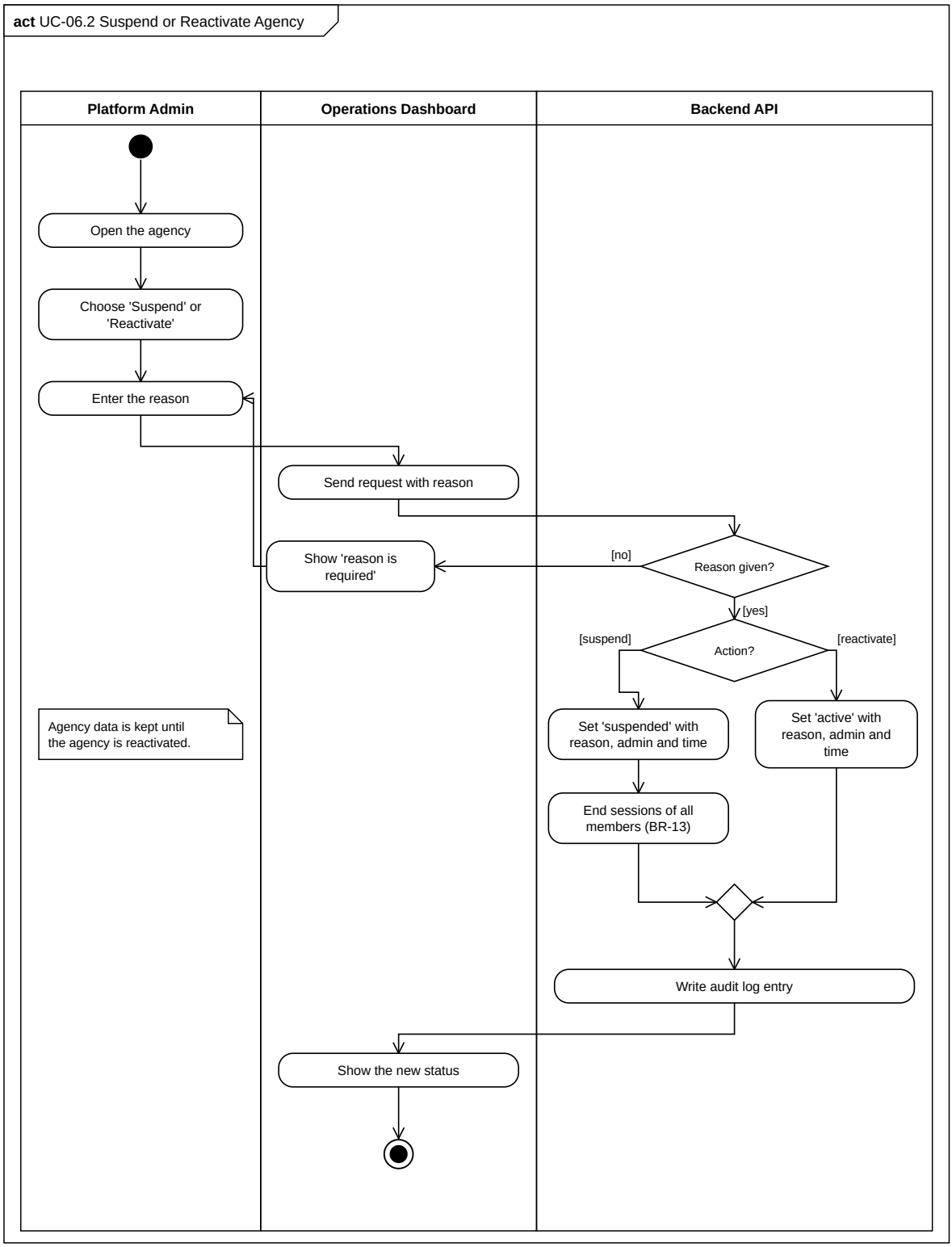
UC-05.9 Manage Units (change price or status) — Sequence diagram



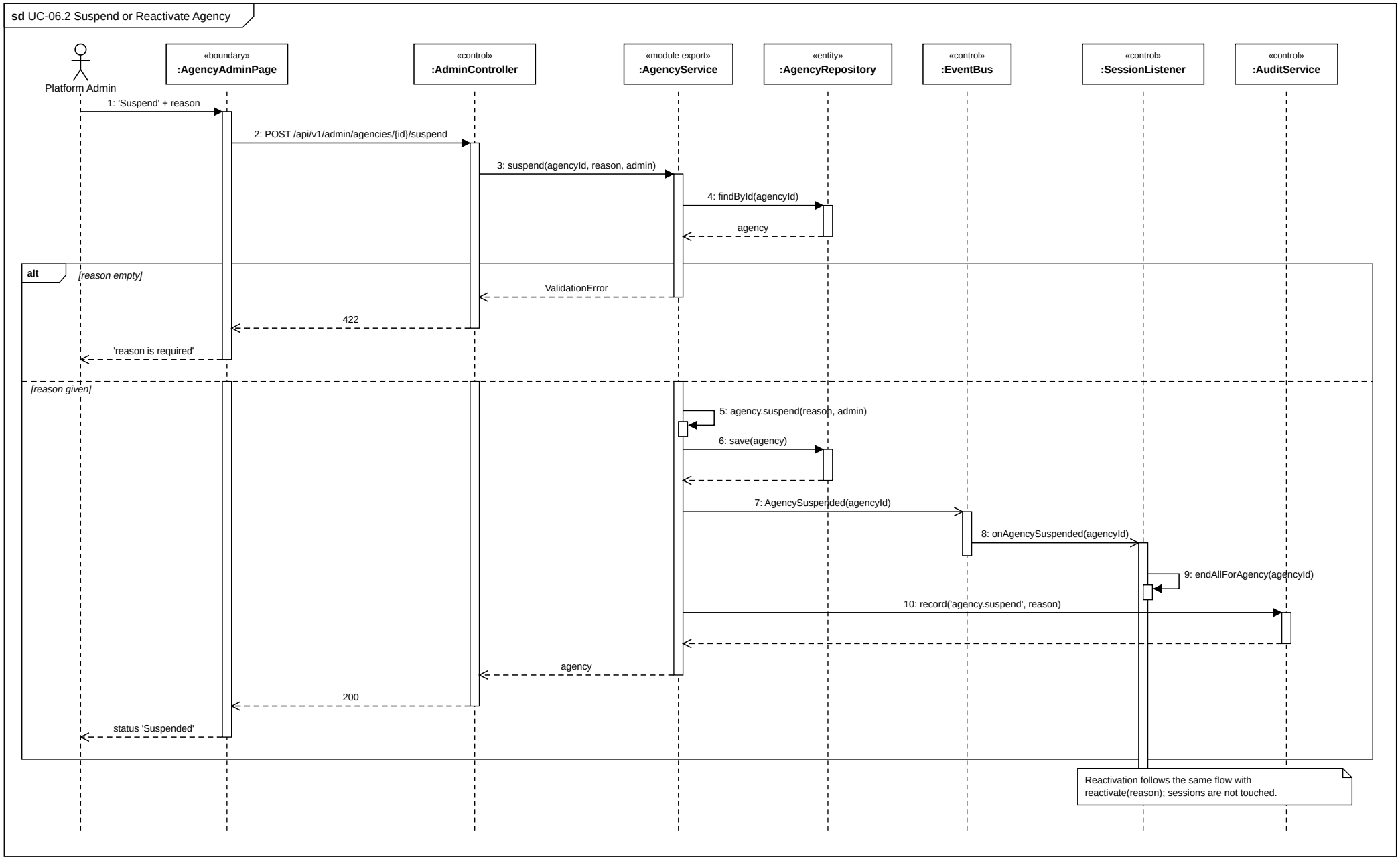
UC-06.2 Suspend or Reactivate Agency

Primary actor	Platform Admin	**Requirements**	FR-ADM-02, FR-ADM-08, BR-13
Description	The Platform Admin suspends or reactivates an agency with a mandatory reason; members of a suspended agency are signed out and cannot log in.		
Precondition	The Platform Admin is logged in.		
Postcondition	The agency status, reason, admin and time are stored; on suspension all member sessions are ended.		

Activity diagram



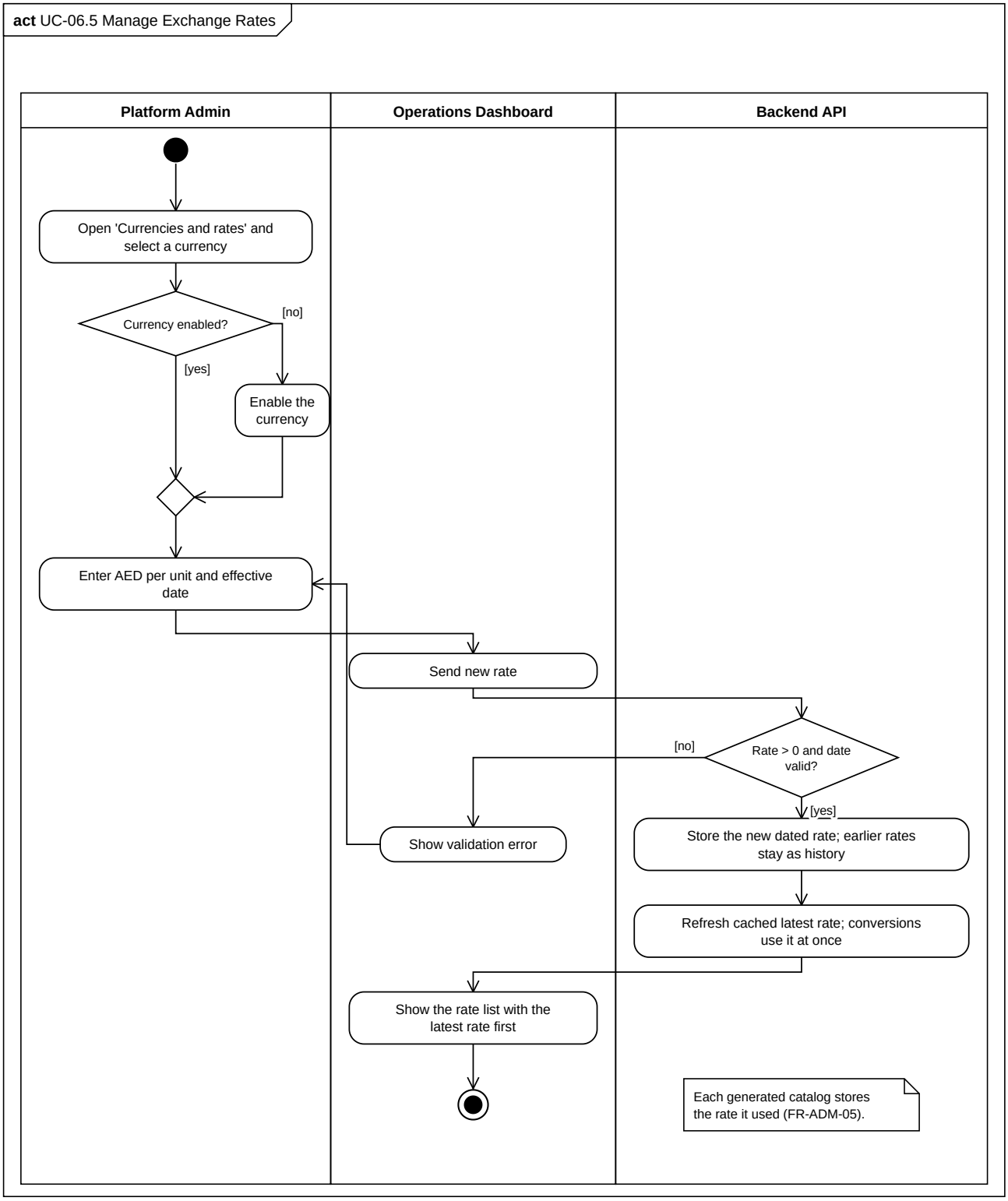
UC-06.2 Suspend or Reactivate Agency — Sequence diagram



UC-06.5 Manage Exchange Rates

Primary actor	Platform Admin	**Requirements**	FR-ADM-05, FR-LOC-01, BR-03
Description	The Platform Admin enables a display currency and enters a dated exchange rate; the latest rate is used for conversions at once.		
Precondition	The Platform Admin is logged in.		
Postcondition	A new dated rate is stored (history kept) and new conversions and catalogs use it.		

Activity diagram



UC-06.5 Manage Exchange Rates — Sequence diagram

